

Assembly Module

Current-state context, functionality and screen inventory

PLPI Batch Assembly Record workflow | Stage 11 | Prepared 17 August 2026

ENTRY	OWNER	NEXT
Allocated assembly room	Assembly Team	Post-Assembly QC

1. Module context

The Assembly module is Stage 11 of the electronic Batch Assembly Record workflow. It receives batches after Production Control has completed verification and room allocation. Its controlled output is a completed, reconciled and electronically signed assembly record ready for Post-Assembly QC.

Workflow position: Pre-Assembly QC -> Production Control / Room Allocation -> Assembly Room -> Post-Assembly QC -> Pre-QP -> QP Release.

2. Business purpose

- Confirm that the correct batch, product, specimen and printed components are present in the allocated room.
- Record the batch start, team briefing and pre-start verification.
- Perform random sample and in-process checks during assembly.
- Reconcile issued, used, damaged and discrepant materials.
- Complete end-of-batch room clearance and retain an electronic audit trail.
- Move the finished batch to Post-Assembly QC only after all four pages are signed.

3. End-to-end user journey

Step	User action	System behaviour
1	Find batch	Search the active queue by B&S batch number or manufacturing lot.
2	Start batch	Confirm start; capture logged-in user and timestamp.
3	Initial checks	Verify BAR, specimen, allocated room and received printed materials.
4	Sample check	Confirm lot and expiry for each box received from Production Control.
5	IPC checks	Complete time-based in-process checks; add extra IPC rows if needed.
6	Reconcile	Record quantities received, used, damaged and discrepant; confirm room clearance.
7	Finish batch	Sign all four pages, record finish audit data and hand off to Post-Assembly QC.

4. Screen inventory

4.1 Assembly batch queue

Shows active and completed batches, batch/lot identifiers, description, quantity, allocated room, status and action. The screen also presents demonstration check-in/check-out/break controls and staff attendance rows.

Figure 1. Assembly Room workflow - active and completed batch queue

The screenshot displays the 'Assembly Room' workflow interface. At the top, a toolbar contains various icons for functions like Machine, PI Label Print, PPM Audit, PPM Locations, PPM Audit History, Regulatory Task, RPI Pack Creation, RPI Approval, Regulatory Log, Product Add, Product, Packing List, Goods-in Summary, Pre OP, Pre Assembly, Batch Checker, BNS Batch Add, and Printer. The main area is titled 'Assembly Room' and includes search fields for 'B&S Batch Number' and 'MFG Lot No'. Below these are two tables: 'Assembly Room Workflow - Active Batches' (4 batches / 1670 packs) and 'Previous Batches - Completed' (5 batches / 2500 packs). The active batches table lists items like 03A1582 (Plutform inhalation) and LVT014 (Lumigan eye drops). The completed batches table lists items like LVT016 (Soludronate oral solution) and LVT017 (Clovate cream). At the bottom, there's a staff attendance table with columns for Check In, Check Out, Break, Username, Start Time, and End Time. The current date is Tuesday, June 16, 2026.

BNS_BATCH_NO	MFG_LOT_NO	DESCRIPTION	QUANTITY	ROOM_NO	STATUS	ACTION
> 03A1582	25086A	Plutform inhalation	650	Room 2	Ready to Start	Start Assembly
03A1582	25086A	Plutform inhalation	350	Room 2	Ready to Start	Start Assembly
LVT014	MFG-LUM-2414	Lumigan eye drops	390	Room 2	Ready to Start	Start Assembly
LVT015	MFG-MAS-2415	Mastical tablets	280	Room 3	Ready to Start	Start Assembly

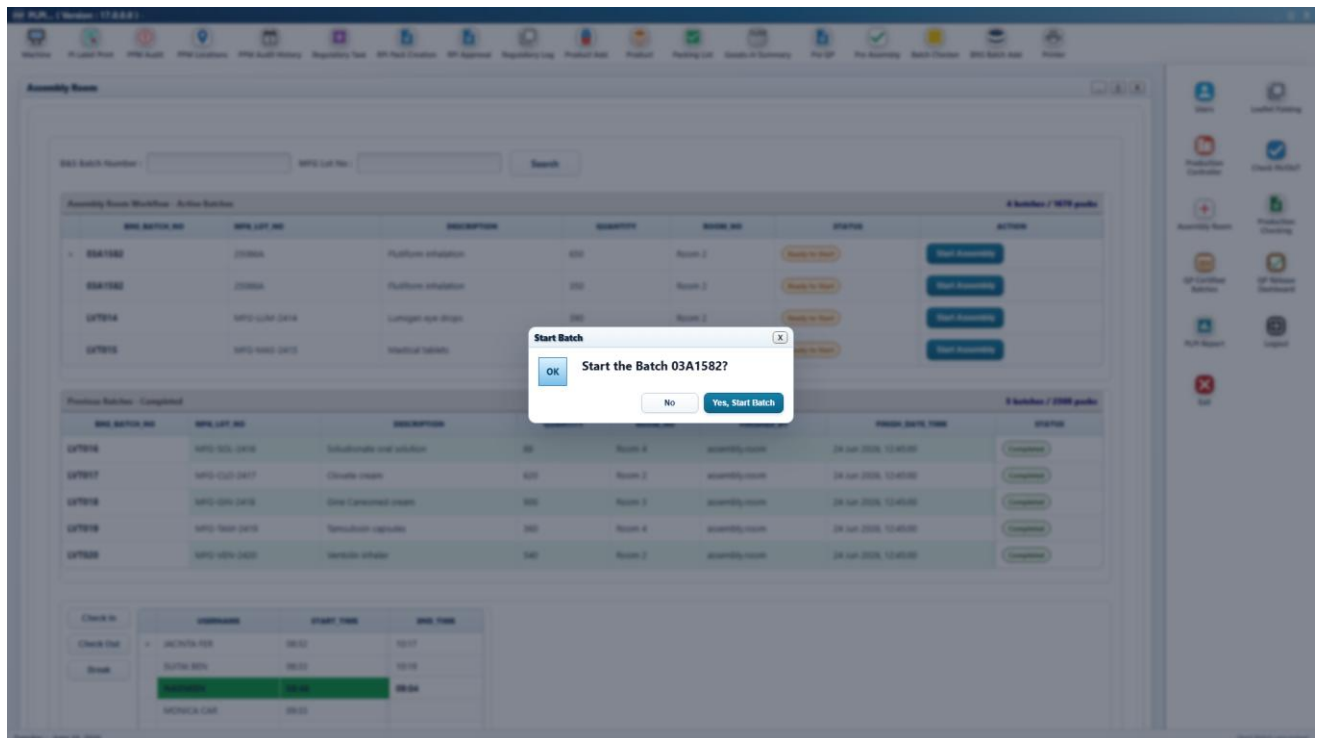
BNS_BATCH_NO	MFG_LOT_NO	DESCRIPTION	QUANTITY	ROOM_NO	FINISHED_BY	FINISH_DATE_TIME	STATUS
LVT016	MFG-SOL-2416	Soludronate oral solution	88	Room 4	assembly.room	24 Jun 2026, 12:45:00	Completed
LVT017	MFG-CLO-2417	Clovate cream	620	Room 2	assembly.room	24 Jun 2026, 12:45:00	Completed
LVT018	MFG-GIN-2418	Gine Canesmed cream	900	Room 3	assembly.room	24 Jun 2026, 12:45:00	Completed
LVT019	MFG-TAM-2419	Tamsulosin capsules	360	Room 4	assembly.room	24 Jun 2026, 12:45:00	Completed
LVT020	MFG-VEN-2420	Ventolin inhaler	540	Room 2	assembly.room	24 Jun 2026, 12:45:00	Completed

CHECK_IN	CHECK_OUT	BREAK	USERNAME	START_TIME	END_TIME
>	JACINTA FER	08:32	10:17		
	SUITAI BEN	08:33	10:19		
	NASNEEN	08:48	09:04		
	MONICA CAR	09:33			

4.2 Start batch confirmation

Opening a batch that has not yet started raises a confirmation dialog. Accepting it records the current user and date/time as the Assembly Room batch start.

Figure 2. Start Batch confirmation and audit message



4.3 Initial Checks

Displays the batch identity strip, allocated room, lifecycle audit fields, briefing field and confirmations for the BAR, specimen and received materials. The page sign-off remains disabled until every required checkbox is selected.

Figure 3. Initial Checks and received-material verification

The screenshot displays the PLPI (Version: 17.0.0.0) software interface for the Assembly Room. The top menu bar includes various functions such as Machine, PL Label Print, PPM Audit, PPM Locations, PPM Audit History, Regulatory Task, RPI Pack Creation, RPI Approval, Regulatory Log, Product Add, Product, Packing List, Goods-in Summary, Pre QP, Pre Assembly, Batch Checker, BNS Batch Add, and Printer.

The main window is titled "Assembly Room" and contains the following sections:

- Scan BarCode Here:** 03A1582
- QTY:** 650
- Table:**

BNS_BATCH_NO	MFG_LOT_NO	DESCRIPTION	QUANTITY	MARKS	BNS_NEW_QTY	ROOM_NO
03A1582	25086A	Flutform inhalation	650	9	650	Room 2
- Buttons:** Partial Start, Partial Finish, Break
- Tabs:** Initial Checks, Random Sample Check, IPC Checks, Reconciliation & Closure
- Initial Checks Section:**
 - Assembly Room: Room 2
 - Batch Started By: assembly.room
 - Batch Start Date / Time: 17 Aug 2026, 15:14:14
 - Batch Finished By: Pending
 - Batch Finish Date / Time: Pending
 - Briefing Done By: assembly.room
 - ☐ Allocated BNS Batch No verified on PLPI against BAR
 - ☐ Specimen verified against BAR
 - ☐ Room leader confirms BAR, printed material and specimen are correct
- Received Materials Section:**
 - ☐ Carton/Braille Label 1: FLU250-10/120/CZ (R4)
 - ☐ Blister Label 1: FLU250-10/120/CZ/B1 (R3)
 - ☐ Blister/Carton Label 2: FLU250-10/120/CZ/B2 (R3)
 - ☐ Leaflet 1: FLU250-10/120/CZ/LT (R5)
 - ☐ Other: Blank Label
- Sign-off Section:**
 - Initial Checks By: assembly.room
 - Date / Time: Pending sign off
 - User Sign Off button

The right sidebar contains navigation icons for Users, Leaflet Folding, Production Controller, Check IN/OUT, Assembly Room, Production Checking, QP Certified Batches, QP Release Dashboard, PLPI Report, Logout, and Exit.

Footer: Tuesday -- June 16, 2026 | Batch 03A1582 started by assembly.room at 17 Aug 2026, 15:14:14

4.4 Random Sample Check

Creates one confirmation row per box passed from Production Control. Manufacturing lot and expiry are populated from the BAR details.

Figure 4. Random sample confirmation by box

RP PLP (Version : 17.0.0.0)

Machine

PI Label Print

PPM Audit

PPM Locations

PPM Audit History

Regulatory Task

RPI Pack Creation

RPI Approval

Regulatory Log

Product Add

Product

Packing List

Goods-in Summary

Pre QP

Pre Assembly

Batch Checker

BNS Batch Add

Printer

Assembly Room

Scan BarCode Here

QTY

03A1582

650

BNS_BATCH_NO	MFG_LOT_NO	DESCRIPTION	QUANTITY	MARKS	BNS_NEW_QTY	ROOM_NO
03A1582	25086A	Plutiform inhalation	650	9	650	Room 2

Partial Start

Partial Finish

Break

Initial Checks

Random Sample Check

IPC Checks

Reconciliation & Closure

Product Random Sample Check

Box Number	Mfg. Lot No.	Exp. Date	Confirmed
Box 01	25088FA	07/2027	☐ Confirmed
Box 02	25088FA	07/2027	☐ Confirmed

2 boxes carried from Production Controller. Mfg. lot and expiry are populated from the BAR details.

Random Sample Check By

Date / Time

assemblyroom

Pending sign off

User Sign Off

Users

Leaflet Folding

Production Controller

Check IN/OUT

Assembly Room

Production Checking

QP Certified Batches

QP Release Dashboard

PLP Report

Logout

Exit

Tuesday - June 16, 2026

Batch 03A1582 started by assembly-room at 17 Aug 2026, 15:14:14

4.5 IPC Checks

Displays the estimated assembly time and minimum IPC count. Each IPC repeats the product identity data needed at the point of checking. Operators can add extra IPC rows.

Figure 5. In-process checks and product identity strip

PPM PLPI... (Version: 17.0.0.0)

Machine P Label Print PPM Audit PPM Locations PPM Audit History Regulatory Task RPI Pack Creation RPI Approval Regulatory Log Product Add Product Packing List Goods-in Summary Pre QP Pre Assembly Batch Checker BNS Batch Add Printer

Assembly Room

Scan BarCode Here QTY
03A1582 650

BNS_BATCH_NO	MPG_LOT_NO	DESCRIPTION	QUANTITY	MARKS	BNS_NEW_QTY	ROOM_NO
03A1582	25086A	Flutiform inhalation	650	9	650	Room 2

Partial Start Partial Finish
Break

Initial Checks Random Sample Check **IPC Checks** Reconciliation & Closure

In Process Checks

Estimated assembly time: 60 minutes Required IPC checks: 3 Add IPC Row

Product	Strength	Pack Size
Flutiform pressurised inhalation, suspension	250/10mcg	120 actuations
ECMA 14/555/12-C	PL No. 18799/2922	Units 1
Mfg. Lot No. 25088FA	Expiry 07/2027	B&S Batch No. 03A1582

☐ IPC 1 ☐ IPC 2 ☐ IPC 3

IPC Checks By Date / Time
assemblyroom Pending sign off User Sign Off

Tuesday - June 16, 2026 Batch 03A1582 started by assemblyroom at 17 Aug 2026, 15:14:14

Users Leaflet Folding
Production Controller Check IN/OUT
Assembly Room Production Checking
QP Certified Batches QP Release Dashboard
PLPI Report Logout
Exit

4.6 Reconciliation and Closure

Captures component quantities received, used, damaged and discrepant. It also records total use, retention sample, yield, damages and comments, followed by end-of-batch room-clearance confirmations.

Figure 6. Material reconciliation and end-of-batch clearance

5. Functional requirements represented in the wireframe

Capability	Current behaviour	Maturity
Queue management	Active/completed queues; room and status visibility; batch and lot filtering	Partially interactive
Lifecycle audit	Start and finish user/date/time records	Interactive
Initial verification	BAR, specimen, room and component confirmation	Interactive with gating
Random samples	One confirmation per allocated box	Interactive with gating
IPC	Calculated minimum checks and additional IPC rows	Interactive
Reconciliation	Received, used, damages, discrepancy, yield and comments	Interactive; discrepancy auto-calculated
Clearance	Four end-of-batch clearance checks	Interactive with gating
Electronic sign-off	User/timestamp per page; page locks after sign-off	Interactive prototype
Handoff	Moves completed batch to Post-Assembly QC	Interactive prototype

6. Rules and completion gates

- The Initial Checks page cannot be signed until all material and pre-start confirmations are complete.

- The Random Sample Check page cannot be signed until every box is confirmed.
- The IPC page cannot be signed until all displayed IPC checks are complete.
- Reconciliation cannot be signed until its required checks and quantity fields are complete.
- Each signed page is locked and displays the signing user and timestamp.
- Signing the fourth page triggers Finish Batch confirmation and the Post-Assembly QC handoff.

7. Current prototype gaps to address in redesign

- Check In, Check Out, Break, Partial Start and Partial Finish are visual controls without implemented actions.
- Barcode scanning is displayed but has no scan validation or error-handling workflow.
- The Search button itself is not wired; filtering currently occurs from input changes.
- IPC photo evidence is described in the module definition but no evidence-upload control appears on the Assembly screen.
- There is no deviation, investigation or supervisor-override workflow for a non-zero discrepancy.
- Electronic sign-off is represented by a confirmation dialog rather than validated re-authentication.
- Prototype records are in memory and do not demonstrate durable storage, versioning or audit export.
- Check-in rows are demonstration data and are not linked to actual attendance actions.
- The reconciliation table pre-populates repeated example quantities across material types and needs component-specific data binding.

8. AI redesign brief

Redesign the Assembly module as a tablet-friendly, GxP-aware workflow for operators working in an allocated room. Preserve traceability and completion gates while reducing visual density and making the current task, outstanding requirements and exceptions immediately visible.

Essential redesign outcomes

- A clear progress stepper for Initial Checks, Samples, IPC and Reconciliation.
- Persistent batch identity, room, quantity, status and active operator context.
- Scan-first batch/component verification with visible match or mismatch feedback.
- Large touch targets and a focused task view suitable for tablets and handheld devices.
- Component-specific reconciliation with live expected balance and prominent discrepancy status.
- Evidence capture attached to the relevant IPC, exception or component row.
- A dedicated exception path with reason, evidence, supervisor review and resolution status.
- Validated electronic signature, audit history and clear locked/read-only states.
- Offline/interruption recovery, draft state and safe resume behaviour.
- Accessible colour contrast with status communicated by text and icon as well as colour.

Suggested prompt for a design AI

Using the supplied Assembly Module reference document and screenshots, redesign the Stage 11 Assembly Room workflow for a regulated pharmaceutical batch-record system. Create a desktop and tablet design with an active-batch queue, scan-first batch start, four-step guided checklist, component-level reconciliation, IPC evidence capture, exception handling, validated electronic signatures and a complete audit trail. Preserve all displayed batch identifiers and completion gates, but simplify hierarchy, reduce dense tables and make incomplete, warning, discrepancy, signed and locked states unambiguous. Produce screen variants for normal flow, mismatch, non-zero discrepancy, evidence upload, signature confirmation and completed handoff.